

Hunter Gotwalt

Atlanta, Georgia • 407-797-8136 • hgotwalt3@gatech.edu • LinkedIn: [Hunter Gotwalt](#) • Projects: [Portfolio](#)

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, Scheller College of Business

Atlanta, Georgia

Master of Science in Quantitative and Computational Finance

August 2025-May 2027

Master of Science in Computational Science Engineering

January 2026-May 2027

- Key Coursework: Probabilistic Models, Numerical Methods, AI for Finance, Computational Statistics, Systems in Computational Finance, Stochastic Process, Derivatives, Optimization, Monte Carlo Methods, Numerical Linear Algebra

FLORIDA STATE UNIVERSITY

Tallahassee, Florida

Bachelor of Science in Applied Mathematics and Statistics

August 2022-May 2025

- Minor: Computer Science
- GPA: 3.9
- Key Coursework: Game Theory, Applied Regression, Deep and Reinforcement Learning, Probabilistic Modeling, Probability Theory, Statistical Inference, Machine Learning, Time Series Analysis, Data Structures and Algorithms

EXPERIENCE

Charles Schwab

Lone Tree, Colorado

Model Risk Oversight - Model Validation Intern

June 2026-August 2026

Booz Allen Hamilton

Panama City, Florida

Data Science/ Software Engineering Intern

June 2025-August 2025

- Collaborated with engineers, focusing on the design, development, and integration of data science and software components
- Developed an Autoencoder using machine learning techniques for real-time anomaly detection
- Integrated a data flow pipeline to support the implementation of satellite communication on an embedded system
- Presented system architecture, design decisions, and performance outcomes to senior engineers and executive leadership

Florida State University

Tallahassee, Florida

Research Assistant

August 2024-December 2024

- Formulated predictive models using machine learning techniques to analyze high-dimensional NFL tracking data
- Performed feature engineering to capture player movement patterns and play outcomes to improve model performance
- Gained experience leading meetings and managing project timelines in a collaborative research setting

PERSONAL PROJECTS

Analyzing Market Uncertainty with Google Trends

- Utilized time series decomposition to filter out the noise of google trends data
- Built a Kalman Filter to identify the current market uncertainty from the search trends
- Composed a risk signal that combined the S&P 500 trade volume and the Kalman derived stress score
- Backtested this strategy, achieving a Sharpe ratio of 1.42 and reducing drawdown to 11% versus 33% for a buy and hold

Market Regime Modeling and Algorithmic Trading Agent

- Segmented financial market regimes using a Hidden Markov Model
- Forecasted future SPY returns with Monte Carlo methods using regime stats and VIX-adjusted volatility
- Engineered a GRU-based predictive model using data from historical returns, volatility signals, and simulated paths, to predict the S&P 500s next day return. Achieved MAE of 0.0102 and RMSE of 0.0140
- Conducted a 4-year backtest, achieving a Sharpe ratio of 0.98 compared to the benchmark's 0.55

SKILLS

Programming: Python, SQL, C/C++, R, SAS

Tools: PyTorch, NumPy, Pandas, Scikit-learn, Git, Linux, cvxpy, statsmodels, SciPy, TensorFlow, Jupyter

Interests: Poker Strategy, Game Theory, Sports Analytics, Golf, Intramural Sports, Coaching Youth Football, Fantasy Football